Chapter 7_ Memory Building AI that LEARNS

Fun fact: our brain, it is processing about 11 million bits of information, yet we are only consciously aware of about 40 to 50 bits. This selective awareness reveals something crucial about how our minds work: memory is not about storing everything; it is about storing what matters.

Emotion tagging helps us prioritize important information and make better decisions.

When we recognize that human memory is more about making meaningful connections than storing perfect records, we can better understand what we should aim for in an AI system: not just more storage but smarter, more contextual ways of using information.

AI memory

LLM is like an echo chamber with limited memory.
The most advanced AI systems can process complex information, but often cannot remember simple details about their users.

Three main goals of an AI agent

- First, **contextual understanding**, a memory-enabled AI system can maintain context across interactions, similar to how we follow a continuous conversation. It is good for problem-solving and interaction.
- Second, **Learning and adaptation**, the best AI implementations are those that can learn from past interactions.
- Third, **personalization at scale**, an AI agent can deliver this by retaining and learning from individual interactions while maintaining privacy and security.

Three layers of an AI agent

1. **Short-term Memory (STM) - immediate context retention**
   It holds recent interactions and ensures contextual continuity within a single session. But, STM has a limited capacity; as new data comes in, older information is overwritten unless transferred to Long-term Memory.
   - **Context Window:** It is a temporary workspace where the AI can see and process information. It is like a whiteboard that cannot hold a limited amount of text, and once it is full, older information must be erased to make room for new incoming information. Normally, the GPT can store up to 100, 000 words or a few hundred thousand tokens. So it is really limited to store all of the information, the AI must decide how to arrange the information to ensure it stays manageable.
   - **Context Ceiling:** the point at which adding more information to an agent's memory starts degrading rather than enhancing performance.
   - Once the information is organized, the **attention mechanism** takes over. It is the AI's

ability to focus on the most relevant parts of the input. It assigns weight to different pieces of information based on our prompt (query).

- Token management: it is an AI's note-taking system, trying to capture the key points without writing everything down. So when the AI almost goes near its memory limit, it summarizes, condenses, and decide wjat to keep in its mental workspace. But this approach has a trade-off, some nuance detail or example might get left out because they seemed less critical at the moment, but later it might become crucial.

Thus, the context window defines its limit, the attention mechanism shapes its focus, and token management enforces trade-offs.

To improve STM:

1. Chunking information: AI works best when its inputs are structured into digestible segments.

2. Priority queuing:** ensuring the most critical information stays accessible in the AI's STM. For example, patients' symptoms and vital signs were prioritized over administrative details.

3. Human oversight

4. Structured and cleared input and desired output

2. **Long-term Memory (LTM) - Structured Retention Over time**

   It can store structured information for future reference. This includes user preference, past interactions, learned workflows, and domain-specific knowledge. LTM enables AI to recognize recurring patterns, recall past interactions, and personalize responses based on accumulated experiences. Unlike STM, LTM is designed to ensure that the AI does not start from scratch in every interaction.

   - **Landmark Attention:** It is a mechanism that equips AI with a sort of mental map to navigate vast amounts of information. Landmark Attention enables AI to divide these massive texts into manageable sections and identify key points, or "landmarks", to focus on. This approach ensures the AI can effectively access relevant details quickly without becoming overwhelmed by the sheer volume of information.

   - **Varied-Size Window Attention (VSA):** it allows the AI to adapt its focus dynamically by offering an adjustable pair of glasses that lets you zoom in on fine details when needed and zoom out to take in the broader context. Creating a varying window based on the requirement.

   - **Retrieval Attention:** it teaches AI to retrieve only the most important pieces of information when needed, much like a person recalling the key point of a conversation without needing to remember every single piece.

   AI with LTM excels at recognizing patterns over time, transforming raw data into actionable insights. Despite the promising performance, there are trade-offs, including optimizing efficiency, protecting privacy, and avoiding overwhelming the model with excessive data.

   To tackle these issues, we are required to give LLMs access to external memory capabilities.

3. **Feedback loops - Learning and adaptation**

   It acts as the self-improvement mechanism of AI memory, refining both STM and LTM over time. By incorporating user feedback, whether explicit(user correction, ratings...) or implicit (engagement patterns, error tracking). The AI adjusts its memory structures to enhance accuracy and relevance. This allows AI to improve continuously by reinforcing useful knowledge and discarding outdated or incorrect information.

- **Reinforcement Learning:** works by rewarding desirable actions, those that achieve a set goal or improve performance, while penalizing failures, just as beneficial traits are "rewarded" in evolution, that being passed on to future generations.
- Feedback loop turns the agent into a living system; it gets smarter as it interacts more with the real world.
  How does the feedback loop system work?
  - Explicit user feedback: Agent asks the user for rating, reviews,...
  - Implicit feedback: the system tracks the user behavior, such as task completion rates, abandoned workflows, or response corrections
    - **System Level Reinforcement signals:** Reward models reinforce good actions (successful task execution) and discourage failure.

To operate the feedback loop, the agents work in a cycle of execution, evaluation, and adjustment:
- Observation: the agent gathers data from interactions (user feedback, task success rates, implicit cues)
- Evaluation: the system assigns a reward score (positive or negative) based on feedback
- Adjustment: The AI agent refines future responses by updating the model's weight, prompt, and workflows
- Deployment: The improved agent is deployed and continues learning over time.
Sometimes, in the process of adjusting or correcting its prediction, it might come across an insightful pattern that does not appeal to humans

The effectiveness of the Feedback loop depends on the complexity of the environment and the quality of the data

Azure or AWS provides a scalable solution for collecting, processing, and analyzing the data at the necessary scale for a complex AI system.
Overfitting is a common limitation where AI systems become too focused on optimizing a specific metric, losing sight of the bigger picture.

Memory landscape of AI

AI memory generally falls into three primary categories:

- **Episodic Memory:** represents the AI's experiential knowledge -- what happened, when, and with whom. This includes conversation histories, user preferences expressed over time, past actions taken, and the outcome observed. It allows the AI to maintain continuity across interactions, even when separated by days or weeks. Redis database works best for storing STM episodic memory, enabling quick access to recent conversations without a complex retrieval mechanism. The storage should store timestamps, user identifiers, interaction summaries, and identified entities/intent.

- **Semantic Memory:** domain knowledge, policy, facts about the world, product catalog, and any other information that exists independently of specific interactions.
    - Robust semantic memory allows agents to provide accurate information without needing to query external systems for every request.
    - We implement vector databases like Pinecone or Weaviate to store semantic embeddings -- numerical representations of concepts, facts, and knowledge that enable similarity-based retrieval. These databases excel at finding conceptually related information, even when exact keyword matches are not present. This memory requires a structured organization with clear categorization, relationships, and metadata to facilitate precise retrieval.
- **Procedural Memory:** adhere to "How to" as a domain involving sequences of actions, decision trees, and workflows that guide the agent through complex tasks. Encoding the troubleshooting procedures into procedural memory allows the agent to walk users through complex processes step by step. = Automate workflow or repetitive tasks
    - These systems must maintain the integrity of process steps, decision logic, and conditional branching.

Implement Pathway: Building Memory from the ground up

Step 1: Select a framework for implementation
Select frameworks that support memory persistence, retrieval, and update mechanisms.
- LangChain: Support all the memory landscapes of AI
- LlamaIndex (Formerly GPT Index): ideal for complex semantic memory requirements.
Step 2: Define memory requirements

- **Episodic memory**, we have to determine:
    - Which user interactions and conversation elements need to persist across sessions
    - How long should different types of episodic information be retained?
    - Which user preferences and behaviors should be tracked over time?
- **Semantic memory**, we have to determine:
    - What domain knowledge is essential for the agent to perform its task
    - Which information sources should populate the knowledge base?
    - Which compliance or regulatory information must be maintained?
- **Procedural memory**, we have to determine:
    - which workflow and processes the agent needs to follow
    - What decision logic governs process selection and execution
    - How strictly processes must be followed versus allowing flexibility

    Step 3: Build Retrieval Mechanisms

1. Analyze the query and current context to generate retrieval cues
2. Uses these cues to fetch relevant memories from each storage system

3. Ranks and filters the retrieved memories based on relevance, recency, and importance
4. Incorporates the most relevant memories into its reasoning process
5. Generates a response that leverages both the current context and retrieved memories
   Step 4: Implement Memory Consolidation
   It is the process that determines what to remember and what to forget.
   - Importance-based consolidation: Evaluate the significance of new information. When a user provides critical details like preferences, requirements, or feedback, our system will flag this information for long-term storage.
   - Frequency-based consolidation: Tracks recurring patterns. Information that appears frequently across interactions is likely important and gets promoted to long-term memory.
   - Explicit consolidation: it occurs when users or the system deliberately mark information for remembrance. "Remember what I prefer for an evening appointment."
   Step 5: Integrate Memory with agent reasoning
   - **Context preparation:** incorporates relevant memories into the context window before the agent generates a response.
     - In-process retrieval: allows the agent to pull additional memories during its reasoning process if it identifies a need for more information.
   - **Post-response consolidation:** evaluates the interaction after a response is generated to identify information for long-term storage.

Keys evolution:

- **Contextual Prioritization Algorithm:** allows the AI system to better determine which memories are most relevant in real-time.
- **Real-time memory consolidation:** the system will analyze and reorganize stored data on the fly.
- **Potential of Multi-agent systems** in improving memory management by distributing the memory load and enhancing collaboration.
- **Meta-learning**: allows AI to memorize relevant experience on the fly without retraining, and learn how to learn more
- **Universal Learner:** the AI agent that can take on any tasks, even without prior training, and achieve mastery through trial, error, and feedback.

Solution for privacy concerns and bias issues

- Transparency is non-negotiable in fostering this trust. The customer or user has to be able to view what the AI remembers, edit or delete specific memories, and set boundaries AI can retain. Build a feature for the user to do so
- Memory Auditing helps identify irrelevant, outdated, or incorrect information and allows users to refine what the AI retains.
- Automated detection of potential privacy risks
- Clear protocols for data retention and deletion

- Patient control over their data through a transparent portal
- To avoid the pitfall, leaders must view data management as a strategic priority.